Front page TIØ4145 2024

Examination paper for TIØ4145 Corporate Finance

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Present at the exam location: No

Permitted examination support material: D **-** No printed or hand-written support material is allowed. A specific basic calculator is allowed.

OTHER INFORMATION

You will find a formula sheet down at the left.

Read the questions carefully and make your own assumptions. Specify in your answer which assumptions you have used as a basis for interpreting/defining the assignment.

The academic person is only contacted in case of errors or insufficiencies in the question set. Address an invigilator if you suspect errors or insufficiencies. Write down the question in advance.

SPECIFIC INFORMATION

Hand drawings:

For questions in Sections 2, 3 and 4, you may choose to answer on handwritten sheets. At the bottom of the question, you will find a seven-digit code. Fill in this code in the top left corner of the sheets you wish to submit.

We recommend that you do this during the exam. If you require access to the codes after the examination time ends, click “Show submission”.

You are responsible for filling in the correct codes on the handwritten sheets. Therefore, read the cover sheet carefully. The Examination Office cannot guarantee that that incorrectly completed sheets will be added to your assignment.

Weighting: Section 1 counts for 20% of the final score, Sections 2 counts for 30%, Section 3 counts for 20%, Sections 4 counts for 30%. Multiple choice questions in Section 1 are graded as follows: 2 points are given for correct answer; 1 point is subtracted for wrong answer; 0 points is given when the question is left unanswered. Only one answer is correct.

Notifications:

Any messages during the exam (e.g., in case of errors in the exam set) will be sent out via notifications in Inspera. A notification will appear as a dialog box on the screen. You can find the notification again by clicking on the bell icon at the top right

Withdrawing from the exam:

If you wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click “Submit blank”. This cannot be undone, even if the test is still open.

Access to your answers:

After the exam, you can find your answers under previous tests in Inspera. Be aware that it may take a working day until any hand-written material is available in the archive.

1.1 Valuation of companies

Which of the following best describes the purpose of the Normalized Earnings Method?
Select one alternative:

- To adjust earnings by excluding non-recurring or extraordinary items in order to reflect a company's sustainable financial performance.
- To forecast future earnings by considering the company's growth potential and market trends.
- To adjust earnings by including anticipated future gains to reflect a more realistic earnings pattern over time.
- To adjust earnings to include all recurring items, regardless of their relevance to core business operations.

Maximum marks: 2

1.2 Bonds

Which of the following statements is FALSE?
Select one alternative:

- The amount of each coupon payment is determined by the coupon rate of the bond.
- Usually the face value of a bond is repaid at maturity.
- Prior to its maturity date, the price of a zero-coupon bond is always greater than its face value.
- Treasury bills are U.S. government bonds with a maturity of up to one year.

Maximum marks: 2

1.3 Risk-Neutral Probabilities

Which of the following statements about the risk neutral probabilities is FALSE?
Select one alternative:

- To ensure that all assets in the risk-neutral world have an expected return equal to the risk-free rate, relative to the true probabilities, the risk-neutral probabilities underweight the bad states and overweight the good states.
- By using the probabilities in the risk-neutral world we can price any derivative security—that is, any security whose payoff depends solely on the prices of other marketed assets.
- After we have constructed the binomial tree and calculated the probabilities in the risk-neutral world, we can use them to price the derivative by simply discounting its expected payoff (using the risk neutral probabilities) at the risk-free rate.
- If all market participants were risk neutral, then all financial assets (including options) would have the same cost of capital—the risk-free rate of interest.

Maximum marks: 2

1.4 Bankruptcy and Financial Distress

Which of the following statements is FALSE?
Select one alternative:

- When a firm fails to make a required payment to debt holders, it is in default
- Bankruptcy is a long and complicated process that imposes both direct and indirect costs on the firm and its investors that the assumption of perfect capital markets ignores.
- Whether paid by the firm or its creditors, the indirect costs of bankruptcy increase the value of the assets that the firm's investors will ultimately receive.
- With perfect capital markets, the risk of bankruptcy in itself is not a disadvantage of debt—bankruptcy simply shifts the ownership of the firm from equity holders to debt holders without changing the total value available to all investors.

Maximum marks: 2

1.5 Market Portfolio

Suppose all possible investment opportunities in the world are limited to the four stocks listed in the table below:

Stock	Price per Share	Number of Shares Outstanding (Millions)
Taggart Transcontinental	$15.60	25
Rearden Metal	$13.00	45
Wyatt Oil	$29.25	10
Nielson Motors	$26.25	26

Suppose you are holding a market portfolio, which includes an investment of $9000 in Rearden Metal. The amount that you have invested in Nielson Motors is closest to:

Select one alternative:

- 7 715
- 9000
- 10 500
- 6 000

Maximum marks: 2

1.6 Real Options

Which of the following statements is FALSE?

Select one alternative:

- In particular, because real options allow a decision maker to choose the most attractive alternative after new information has been learned, the presence of real options adds value to an investment opportunity.
- To make an investment decision correctly, the value of embedded real options must be included in the decision-making process.
- When you do not have the option to wait, it is optimal to invest in any positive-NPV project.
- A key distinction between a real option and a financial option is that real options, and the underlying assets on which they are based, are often traded in financial markets.

Maximum marks: 2

1.7 Market efficiency

Why do actively managed funds often fail to generate positive alphas on a consistent basis?

Select one alternative:

- Most fund managers lack the skills necessary to outperform the market
- Competition among fund managers and market efficiency reduce opportunities for excess returns
- Fund managers prioritize diversification over returns, which limits alpha generation
- Fund managers face limitations in accessing timely information compared to individual investors

Maximum marks: 2

1.8 Agency Costs of Leverage

Which of the following statements is FALSE?

Select one alternative:

- When a firm faces financial distress, it is the creditors of the firm who gain from sufficiently risky investments, even if they have negative NPV.
- Agency costs are costs that arise when there are conflicts of interest between stakeholders.
- In some circumstances, managers may take actions that benefit shareholders but harm the firm's creditors and lower the total value of the firm.
- When a firm has leverage, a conflict of interest exists if investment decisions have different consequences for the value of equity and the value of debt.

Maximum marks: 2

1.9 CAPM

Suppose that the risk-free rate is 5% and the market portfolio has an expected return of 13% with a volatility of 18%. Monsters Inc. has a 24% volatility and a correlation with the market of .60. Assume the CAPM assumptions hold.

Monsters' beta with the market is closest to:

Select one alternative:

- 1.3
- 0.6
- 1.0
- 0.8

Maximum marks: 2

1.10 Sustainable finance

What is "double materiality" in the context of corporate reporting and sustainability?

Select one alternative:

- The focus on balancing financial performance with non-financial goals related to internal operations, like efficiency and workforce diversity
- The consideration of only impact materiality, which focuses on the environmental and social impacts of the company on society
- The obligation for companies to disclose all material information affecting stakeholders and shareholders quarterly
- The consideration of both financial materiality, which focuses on risks and opportunities impacting the company's financial performance, and impact materiality, which focuses on the company's environmental and social impacts on society

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Maximum marks: 2

2.1 Question 2.1 (15%)

You are asked to price options on OceanMotion Corporation's stock. OceanMotion's stock price can go up by 15 percent every year, or down by 10 percent. Both outcomes are equally likely. Assume that OceanMotion does not pay dividends. The risk-free rate is 5%, and the current stock price is $100.

i) Price a European put option on OceanMotion with a maturity of 2 years and a strike price of $100.

ii) Suppose you expect the price of OceanMotion's stock to have little variance in the future. How could you design a strategy (using options) to take advantage of this?

Fill in your answer here

Format

Words: 0

Maximum marks: 15

2.2 Question 2.2 (15%)

Disregard the Question 2.1. Imagine now that you are the CFO of a Norwegian aquaculture company. In this position, you are currently analyzing a project to develop a new technology for salmon farming designed to control sea lice. Based on today's market conditions, the project is estimated to generate cash inflows with a present value of 300 million NOK. The development of this lice-control technology will cost 250 million NOK.

However, salmon prices fluctuate significantly, with an estimated annual volatility of 15%, which will also affect the cash inflows from this project. In order to launch this technology, you need a government permit. You have been granted a permit for a period of 3 years, after which it will expire if the technology has not been developed.

Assume the annual risk-free interest rate is 5%.

i) What is the value of the project based on the NPV approach? What would be your decision based on the NPV rule?

ii) Calculate the value of the project as an option using the Black-Scholes option pricing model. Here, make the simplifying assumption and interpret the project as a European style option.

iii) Is the value you found in ii) different from NPV you found in i)? Why?

Fill in your answer here

Format

Words: 0/200

Maximum marks: 15

3.1 Question 3.1 (6%)

EcoNor Ventures, an ambitious company leading the green transition, has assets with a market value of $600 million, $70 million of which are cash. The company has debt totaling $250 million and currently has 7 million shares outstanding. Assume perfect capital markets.

i) What is EcoNor's current stock price?

ii) If EcoNor distributes the cash of $70 million as a share repurchase, what will its stock price be after the share repurchase?

ii) How many shares will remain outstanding after the repurchase?

Fill in your answer here

Format

Words: 0/200

Maximum marks: 6

3.2 Question 3.1 (6%)

Now assume that EcoNor distributes the $70 million as a dividend instead of repurchasing shares. Keep all other assumptions from Question 3.1 the same.

i) On the ex-dividend date, what should happen to EcoNor's share price, and why?

ii) Explain how an investor could earn a profit in two scenarios:

1. If the stock price changed less than the amount you found in i).
2. If the stock price changed more than the amount you found in i).

Fill in your answer here

Format

Words: 0

Maximum marks: 6

3.3 Question 3.3 (8%)

As we have seen, EcoNor has the option to distribute excess cash as dividends or to use it to repurchase shares.

Would investors prefer one option over the other if they prioritize receiving cash? Explain your answer by considering an investor who owns 100 shares of EcoNor and prefers cash over additional shares. Keep the assumptions about EcoNor from Question 3.1. Assume in addition that the investor retains all 100 shares after the share repurchase.

Fill in your answer here

Format

Words: 0

Maximum marks: 8

4.1 Question 4.1 (10%)

Suppose *Grizzly Kicks and Boots* (GKB) has earnings per share of $2.30 and EBITDA of $30.7 million. GKB also has 5.4 million shares outstanding and debt of $125 million (net of cash).

You believe *Rugged Outdoors Inc.* is comparable to GKB in terms of its underlying business, but Rugged Outdoors has little debt. Rugged Outdoors has a P/E of 13.3 and an enterprise value to EBITDA multiple of 7.4.

i) Estimate the value of GKB's shares using P/E and enterprise value to EBITDA multiples.

ii) Which estimate is likely to be more accurate and why?

iii) Name two limitations of using multiples for valuation

Fill in your answer here

Format

Words: 0

Maximum marks: 10

4.2 Question 4.2 (5%)

Conduct a sensitivity analysis on the P/E multiple used to value Grizzly Kicks and Boots (GKB). Specifically, assume that the P/E ratio for Rugged Outdoors Inc. could vary by ±10%. Keep other assumptions from Question 4.1.

i) Calculate the impact on GKB's estimated share value if the P/E multiple of Rugged Outdoors changes to 14.63 (+10%) and 11.97 (-10%).

ii) Discuss how sensitive GKB's valuation is to changes in the P/E multiple and what this suggests about the reliability of the P/E-based valuation in this context.

Fill in your answer here

Format

Words: 0

Maximum marks: 5

4.3 Question 4.3 (15%)

Disregard Questions 4.1 and 4.2. Consider now Xpertise AI, a company claiming to have mastered everything from social media to space exploration algorithms. It is expected to generate a free cash flow of $8 million in the coming year, with this cash flow projected to grow at 3% annually. Xpertise AI has an equity cost of capital of 13%, a debt cost of capital of 7%, and faces a 21% corporate tax rate. The company also maintains a constant debt-to-equity ratio of 0.5.

i) Calculate Xpertise AI's Weighted Average Cost of Capital (WACC).

ii) Determine the value of Xpertise AI as an unlevered company, ignoring the effects of debt.

iii) Calculate the value of Xpertise AI's interest tax shield.

Fill in your answer here

Format

Words: 0

Maximum marks: 15